Machine Learning Algorithms and Applications

Lecture 1

(Block 1)

UTS
UNIVERSITY OF TECHNOLOGY SYDNEY

Teaching Staff

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Author
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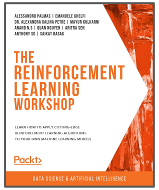
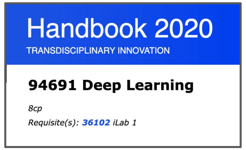
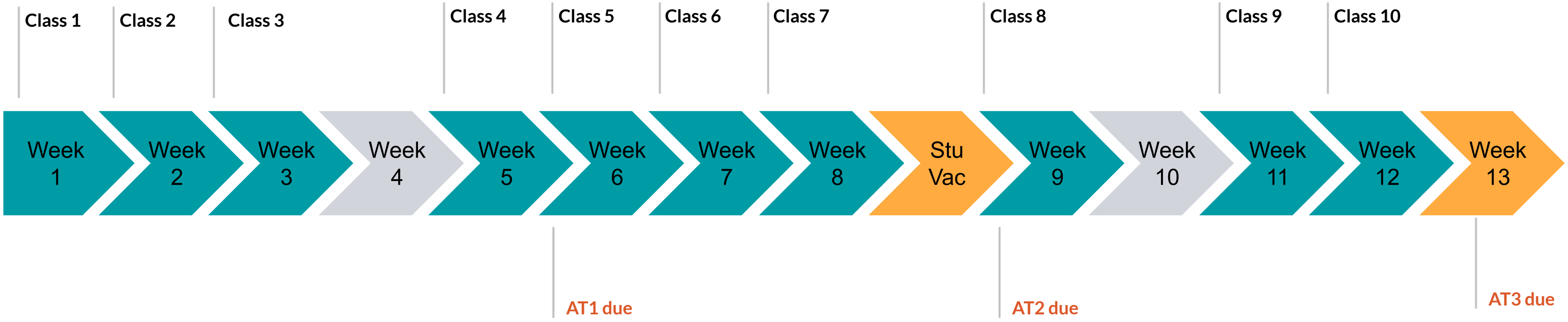
- **2016**
  - Process Analyst at Fairfax Media
  - Enrolled in MDSI
  - First Prize at Unearthed Hackathon Sydney, Pepper Money Hackathon and GovHack Sydney
- **2017**
  - Completed 1st year of MDSI (full-time)
  - Joined Hyperanna as Head of Data Science
- **2018**
  - 2nd year of MDSI (part-time)
- **2019**
  - Graduated from MDSI
  - Joined Douugh as Head of AI & Data Science
  - Author of *The Data Science Workshop* (Packt)
- **2020**
  - Lecturer of Deep Learning elective (UTS), Supervisor of iLab1/2 (UTS)
  - Author of *The Deep Learning Workshop*, *The Applied AI Workshop*, *The Reinforcement Learning Workshop* (Packt)
  - Joined OnlinePajak as Head of Data
  - Lecturer of Applied/Advanced DS for Innovation (UTS Open)
- **2021**
  - Lecturer of Big Data Engineering , Data Science Practise electives, Supervisor of iLab1/2 (UTS)
  - Author of *The Tensorflow Workshop* (Packt)




Handbook 2020
TRANSDISCIPLINARY INNOVATION
94691 Deep Learning
8cp
Requisite(s): 36102 iLab 1

2

Weekly Planning



Integrity Expected by Students

Importance of Integrity

- Integrity fosters a trustworthy and respectful learning environment.
- It enhances personal growth, character development, and professional reputation.
- Demonstrating integrity cultivates a sense of responsibility and accountability.

Academic Integrity

- Adhering to the principles of honesty and fairness in all academic activities.
- Avoiding plagiarism, cheating, or any form of academic dishonesty.
- Citing and referencing sources accurately to give credit to others' work.

Respect for Others

- Valuing diverse perspectives, opinions, and experiences of fellow students.
- Engaging in respectful and constructive discussions.
- Treating others with dignity, fairness, and empathy.

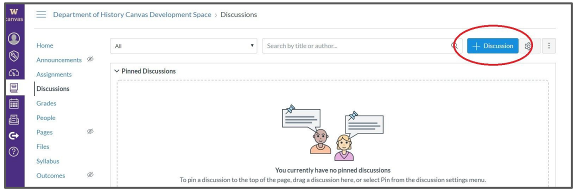
Responsibility and Accountability

- Meeting deadlines and commitments.
- Taking ownership of one's actions and consequences.
- Reporting any observed academic misconduct to maintain the integrity of the community.

Questions

- **Lecture-related**: create a new discussion with prefix "[LECTURE 1] <description>"
- **Labs-related**: create a new discussion with prefix "[LAB 1.1] <description>"
- **Assignment-related**: create a new discussion with prefix "[ASSIGNMENT 1] <description>"
- **Technical issues**: create a new discussion with prefix "[TECHNICAL] <description>"

Canvas Discussion



- For Personal Question **ONLY**

Email to: anthony.so@uts.edu.au

UPASS

- Free study sessions with trained and super friendly senior student leaders
- See www.uts.edu.au/upass-sessions

"Bring all your dumb questions! No one judges." (student, Spring 2022 semester)

Questions:

www.tinyurl.com/upassfilm

upass@uts.edu.au

UPASS Sessions (starting Week 3)
with **Katherin Gomez Londono**
on
Thursdays 4-5pm CB01.05.01
Fridays 4-5pm CB.08.03.005

Learning Objectives

1. Understand the different elements of Machine Learning
2. Manage a Machine Learning project end-end
3. Analyse dataset and propose relevant approaches
4. Assess model performance and make recommendation
5. Train Machine Learning models
6. Communicate and present results

7

Assignments

Assignment	Details	Submission	Due Date	Weight
1	Regression Models	Individual	29/03/2025 11:59pm	30%
2	Building and Interpreting a classification model	Individual	26/04/2025 11:59pm	30%
3	End-end Data Science Project	Group	24/05/2025 11:59pm	40%

MLAA

Learning Experience

Learn by doing

Hands-on approach

Gradual learning curves

Real problems

Real world dataset and business problems

Best practices from industry

Human-centered approach

Deal with uncertainty

Ethics and integrity

Responsible AI

Innovation

Problem solving

Challenges

Creative thinking

Plan for Today

- Introduction to DS and ML
- Regression Problem
- CRISP-DM and Agile Methodology
- Sklearn
- Lab

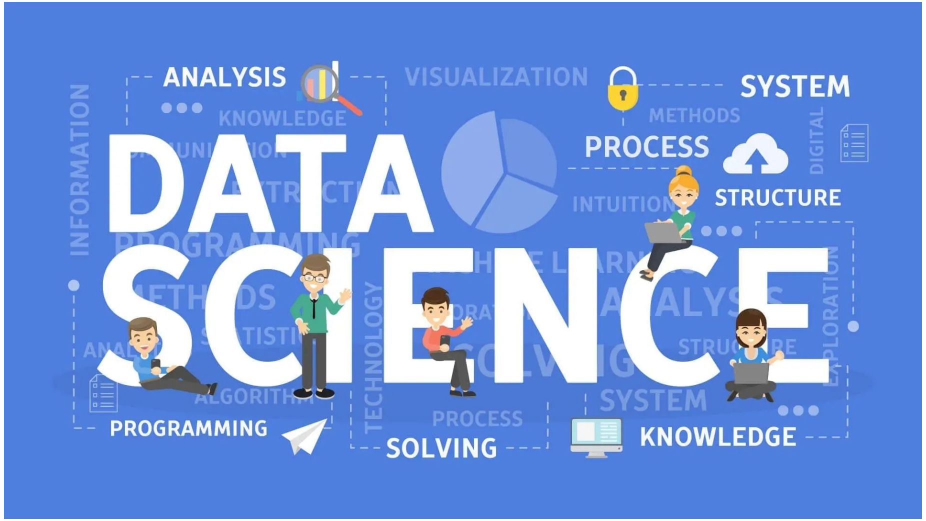
10

Introduction to Data Science

Definition of Data Science

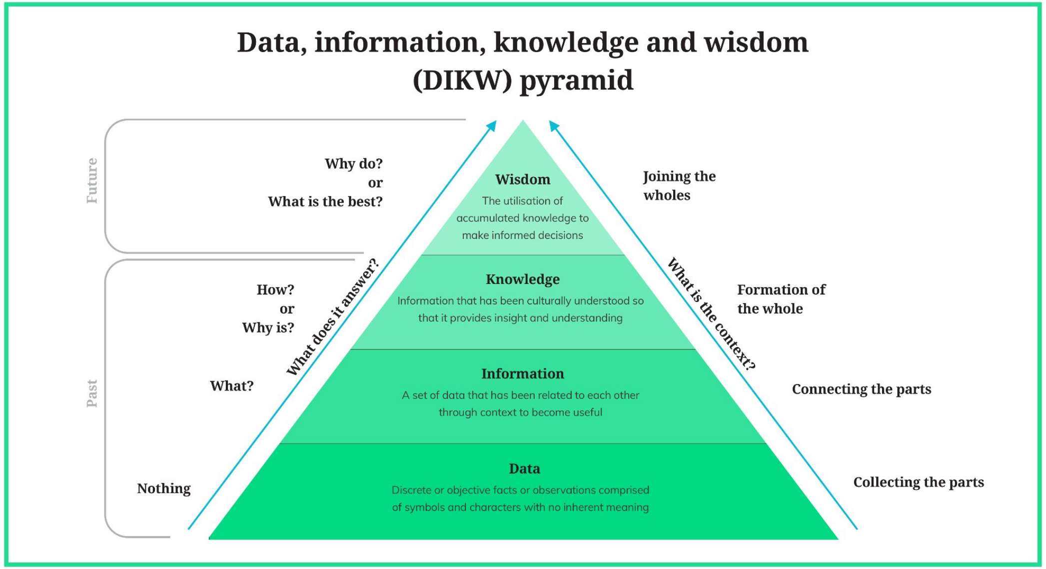
“Data science is an **interdisciplinary** field focused on **extracting knowledge** from typically large **data sets** and applying the knowledge and insights from that data to **solve problems** in a wide range of application domains.”

Wikipedia

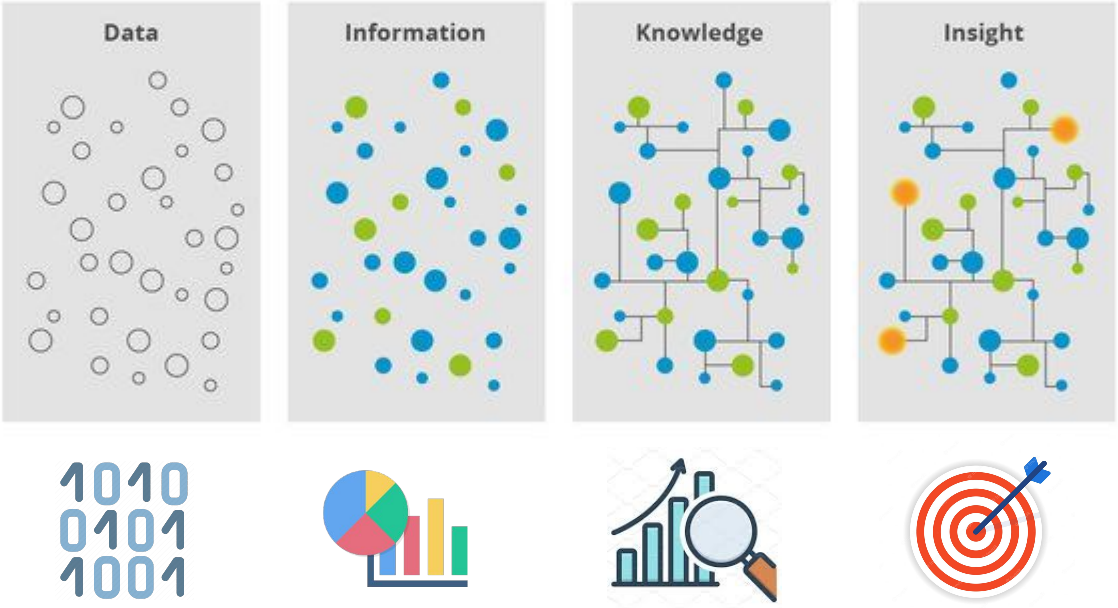


From Data to Decision

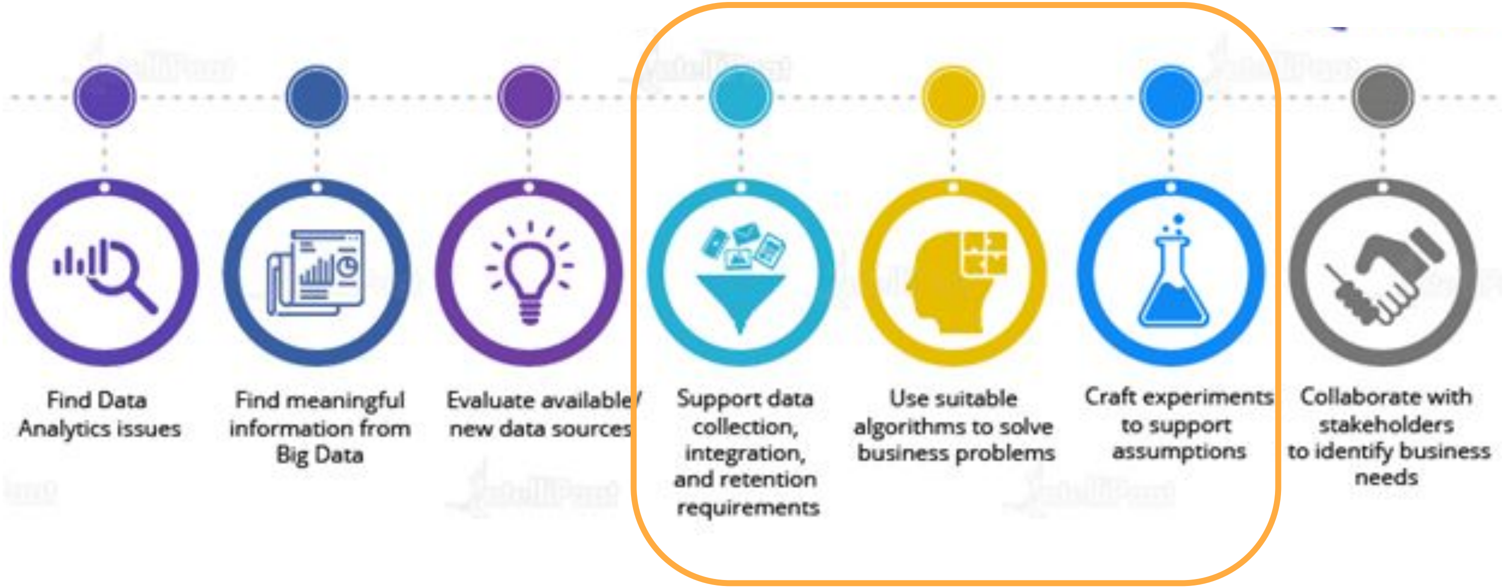
It is about formulating data science problems collecting, preparing, analysing data, presenting findings to inform high-level decisions in a broad range of application domains.



From Data to Actionable Decision



Expectations from a Data Scientist

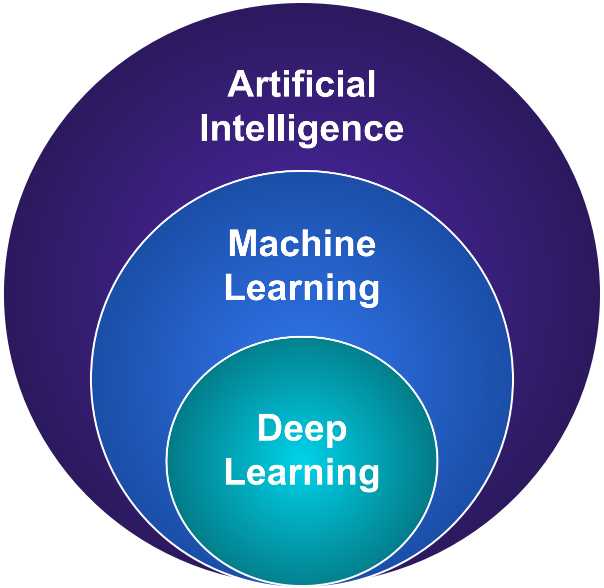


Introduction to Machine Learning

What is Machine Learning

“Machine learning (ML) is the study of computer algorithms that improve automatically through experience.It is seen as a subset of artificial intelligence. Machine learning algorithms build a mathematical model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to do so.”

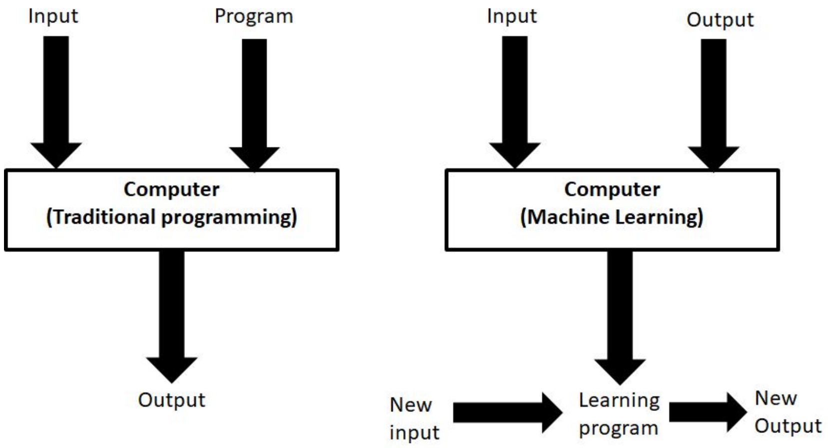
Wikipedia



Machine Learning, a new paradigm

"Machine learning (ML) is the study of computer algorithms that **improve automatically** through **experience**.It is seen as a subset of artificial intelligence. Machine learning algorithms build a **mathematical model** based on sample data, known as "training data", in order to make predictions or decisions **without being explicitly programmed** to do so."

Wikipedia



Applications of ML

ML can be applied in any industry as long as you have data.

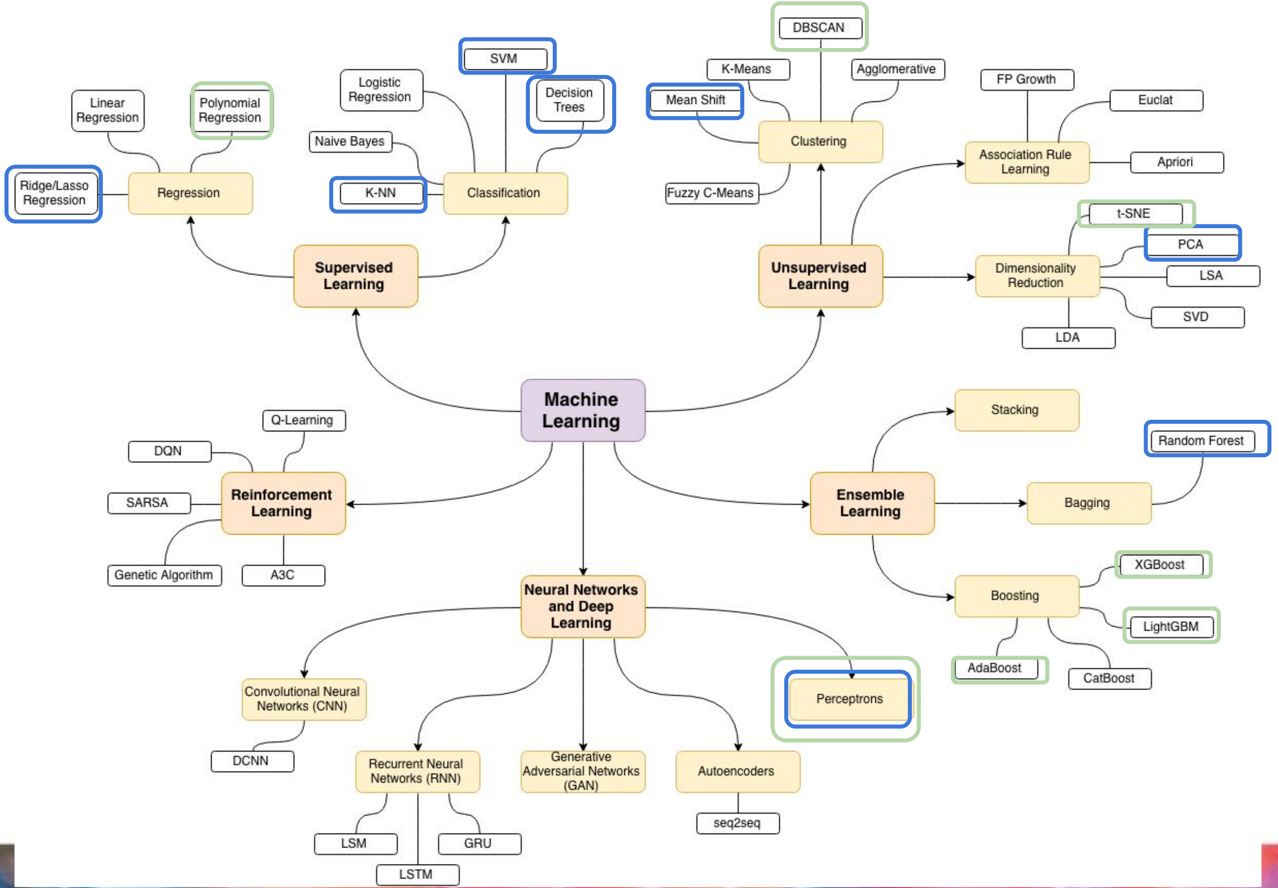
Examples are:

- Finance
- Manufacturing
- Retail
- HR
- Healthcare
- Education
- …

Use Cases


facebook
Social Analytics
amazon
Improving E-commerce experience
UBER
Optimizing Rides
Bank of America
Increasing Customer experience
Airbnb
Improving Searches
Spotify
Music Recommendation

ML Algorithms



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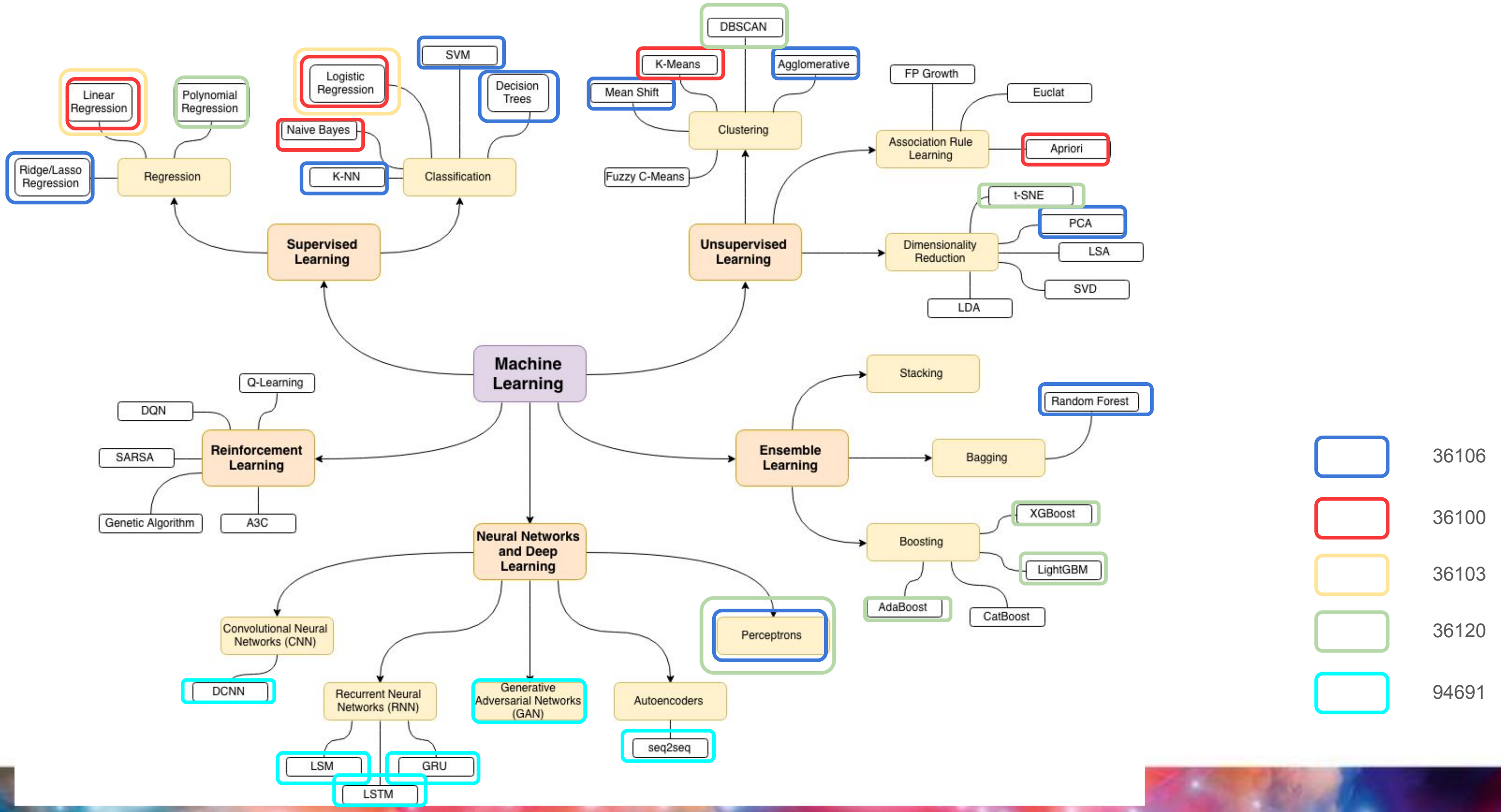
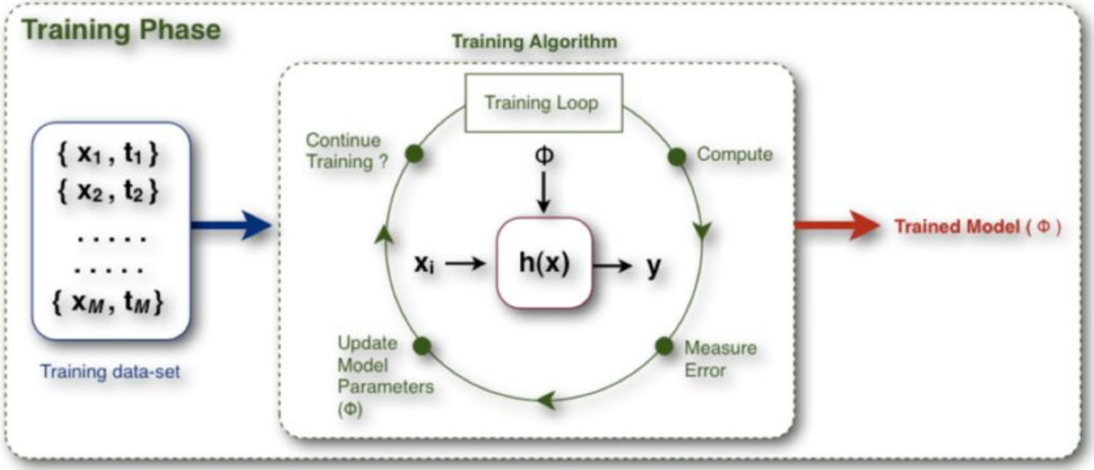
ML Algorithms



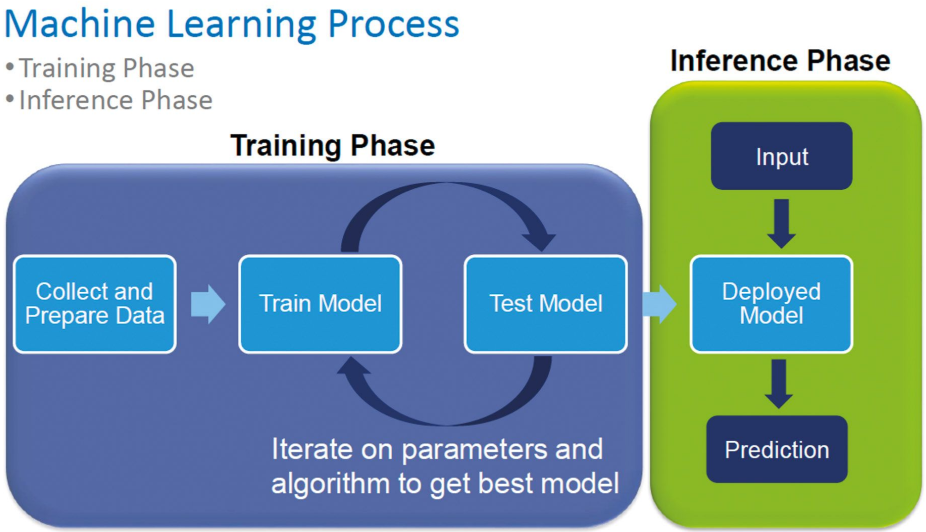
Learning Process

At a high-level, the goal of training a model is to find the best mathematical function that will produce the most accurate results.



X refers to features or independent variables or inputs
Y refers to a target variable or output
H refers to the mathematical function

Training versus Inference



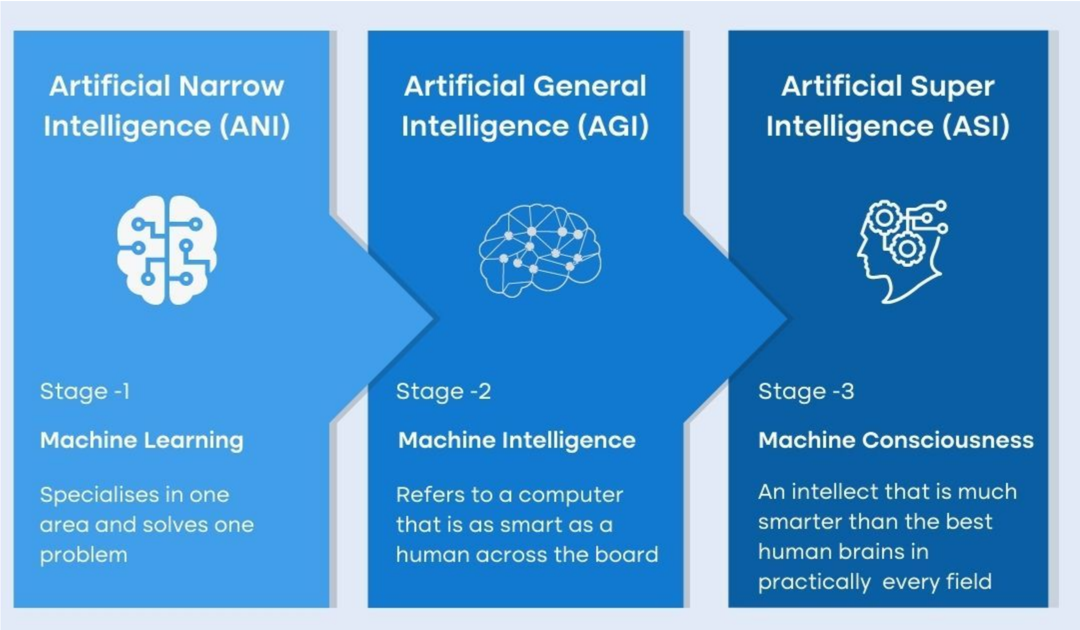
- **Training**: ML model learns relevant patterns to the business objective from the prepared dataset (experimentation phase). The goal is to find the best values for its internal parameters

- **Inference**: ML model applies its learning to new unseen inputs (production). Values for its internal parameters are not updated anymore.

Types of Learning

Criteria	Supervised ML	Unsupervised ML	Reinforcement ML
Definition	Learns by using labelled data	Trained using unlabelled data without any guidance.	Works on interacting with the environment
Type of data	Labelled data	Unlabelled data	No – predefined data
Type of problems	Regression and classification	Association and Clustering	Exploitation or Exploration
Supervision	Extra supervision	No supervision	No supervision
Algorithms	Linear Regression, Logistic Regression, SVM, KNN etc.	K – Means, C – Means, Apriori	Q – Learning, SARSA
Aim	Calculate outcomes	Discover underlying patterns	Learn a series of action
Application	Risk Evaluation, Forecast Sales	Recommendation System, Anomaly Detection	Self Driving Cars, Gaming, Healthcare

3 Stages of Artificial intelligence

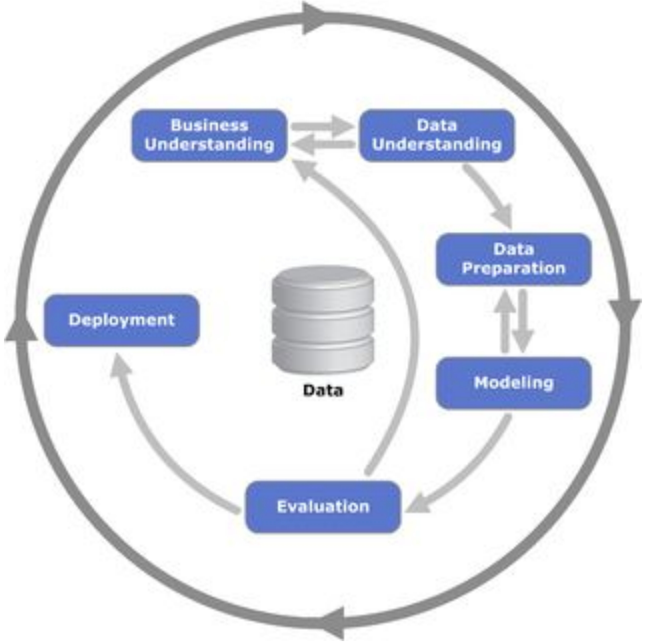


CRISP-DM
and
Agile Methodology

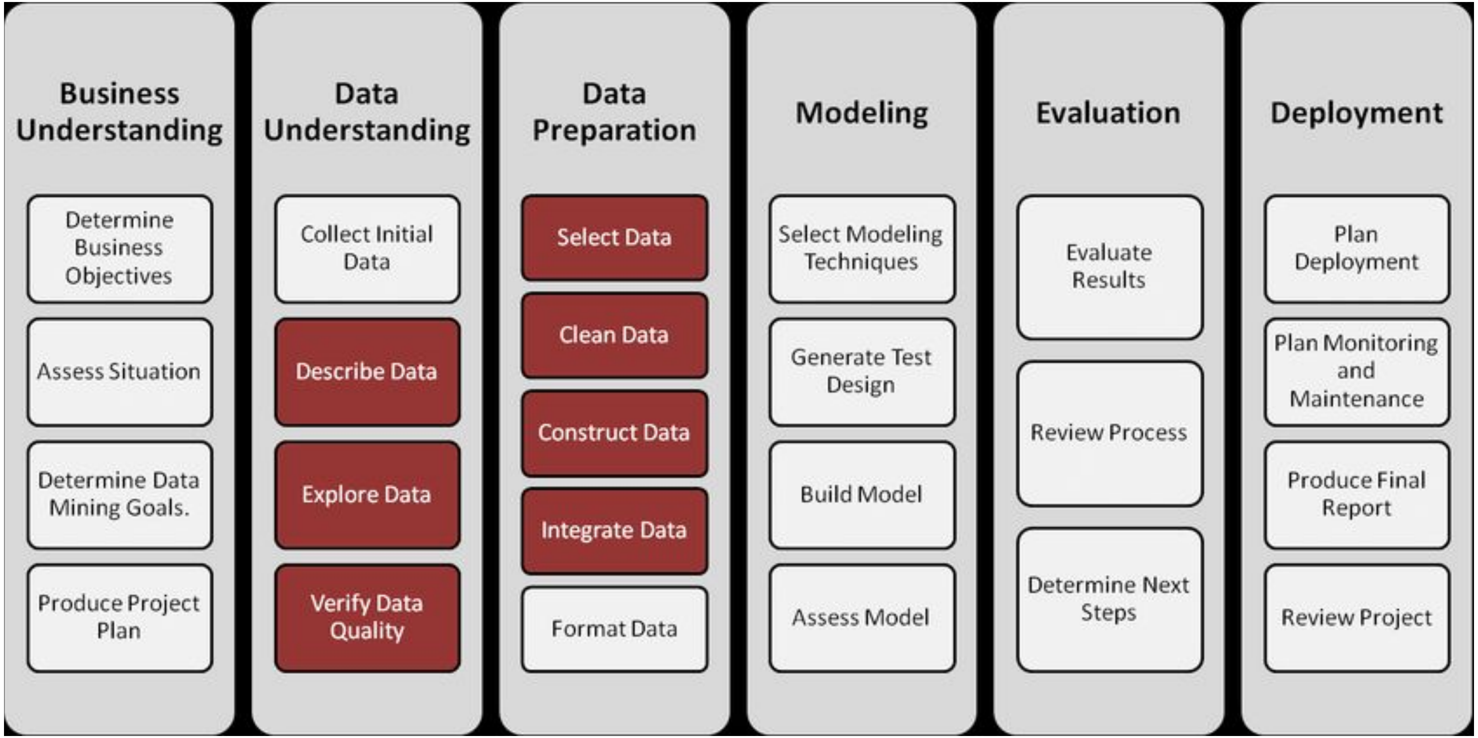
CRISP-DM

Stands for **cr**oss-**i**ndu**s**try **p**rocess for **d**ata **m**ining.

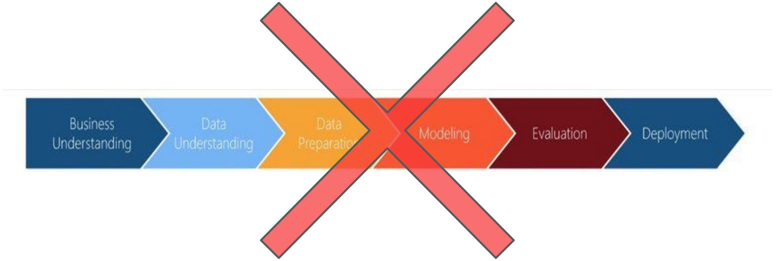
It is a methodology that provides a structured approach to manage a data science project.



6 Steps of CRISP-DM



Not Just a Sequence of Tasks



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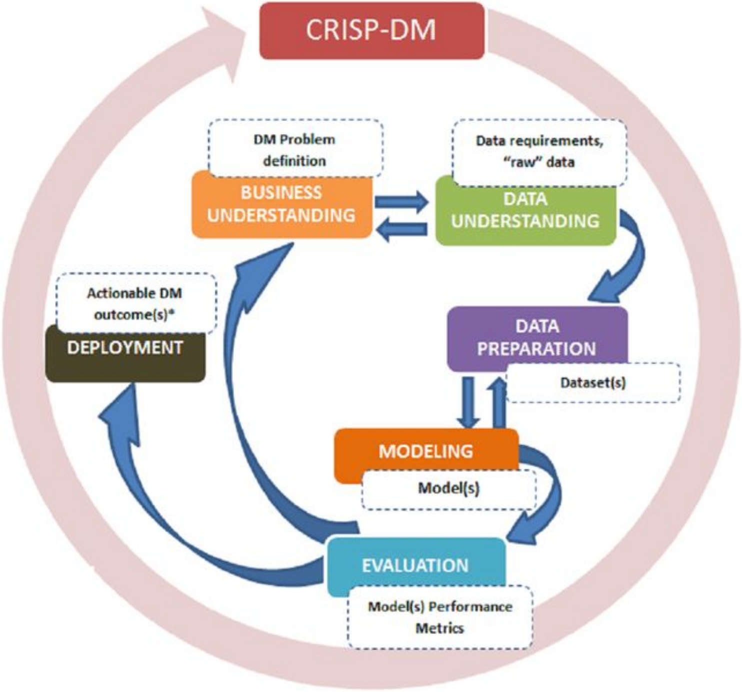


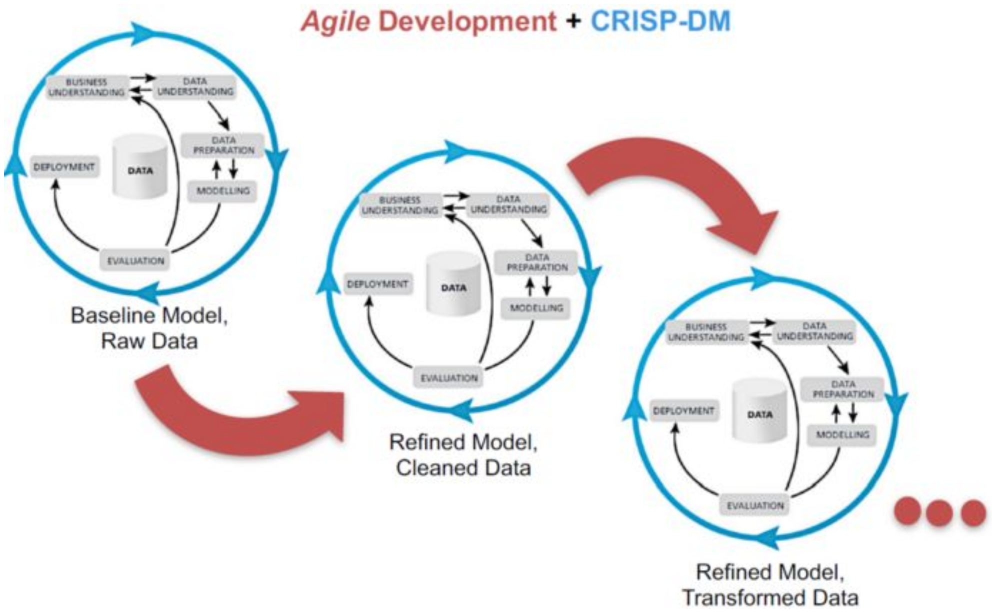
Agile Approach

Why Science in Data Science?

- Applying a scientific approach to solve a problem with a high level of uncertainty

At the beginning of a project, can you estimate:

- How much time will you need?
- How much data cleaning will you perform
- How many new features will you create
- How many algorithms will you use
- How many hyperparameters will you tune?
- Will you even be able to achieve the business objective?



Experimentation

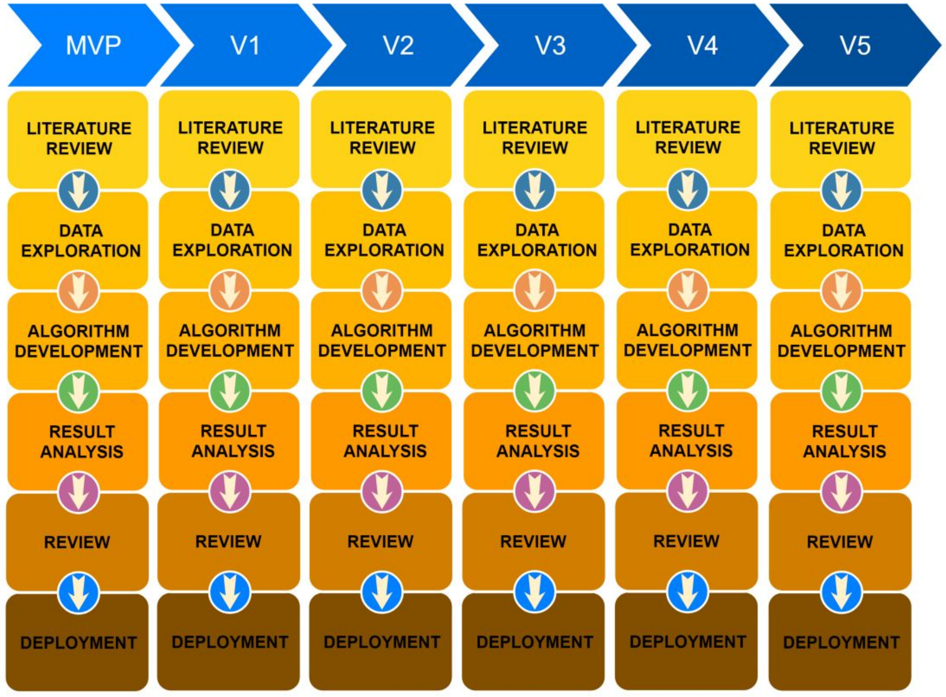
When facing a problem with a high level of uncertainty, you can’t plan upfront and manage a project in delivery mode.

You need to turn the project into a exploratory exercise by running experiments, analysing results, using learning to plan for the next experiments.

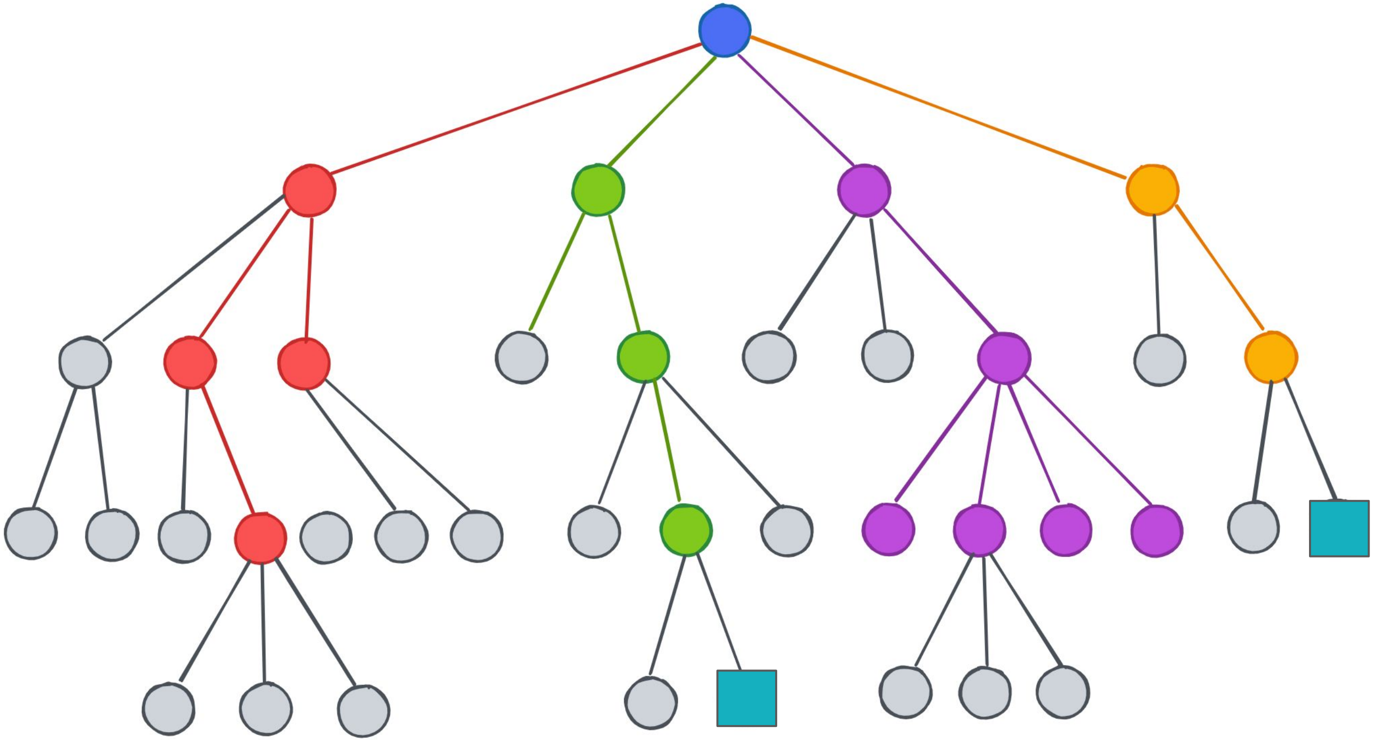
You need to define an hypothesis you want to test for each experiment. At the end you want to confirm if this hypothesis was correct or not.

Experiment after experiment, your hypotheses will become more and more specific.

[Link to Experiment Report]()



Navigating Through Uncertainty



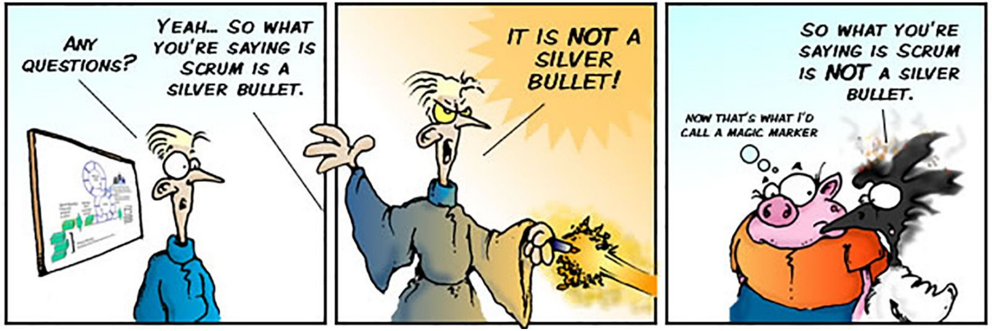
Be mindful

Running Agile doesn't mean:

- You will achieve the set business objectives every single time
- There is one single solution, there may be multiple

Agile means:

- Try multiple approaches
- Get learnings for every single experiment
- "Fail" (learn) fast (business should be grateful)
- Time-boxing your experiments
- Focus on what is important



Introduction to Regression Problem

What is a Regression Task

Supervised learning

Requires a target variable containing the true values.

Continuous Variable

The target variable can take an infinite possible values within a range

Objective

Find a model that will approximate the correlation between the target and features

Regression Application

(Retail) Predict volume of sales per day:
between \$0 and \$10M

(Manufacturing) Predict next equipment failure:
between 1h to 33,800h

(Finance) Predict stock price:
between \$0.01 and \$5000.00

(Medical) Predict number of patients in emergency:
between 0 to 20,000

...

Regression Example

Predicting Stock Price

Target | Features

Price	Industry	Average in last 7 days	Spread in last 7 days	Traded Volume	Turnover
35	IT	32	3	120732	1M
40	Retail	46	10	45234	3M
200	Financial Services	205	1	2464	1B
140	Medical	120	15	43757	500M

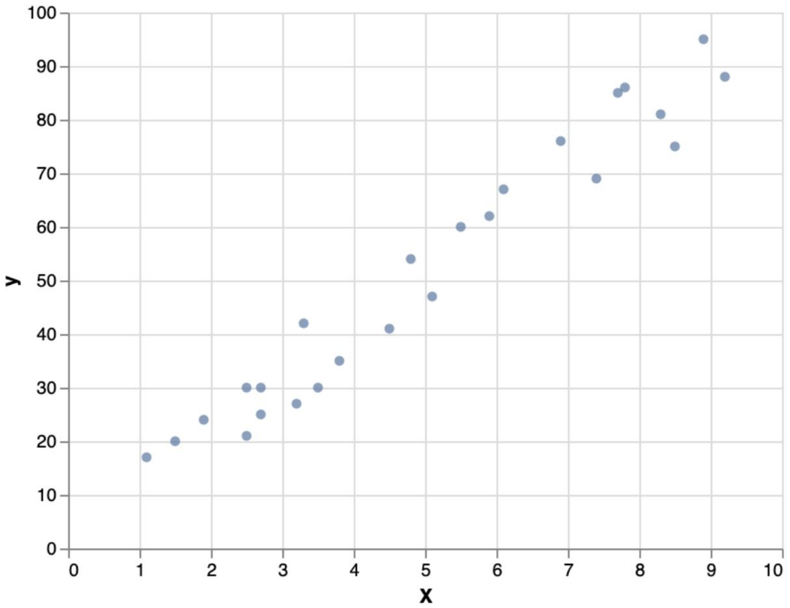
Model Parameter vs Hyperparameter

Example: Univariate Linear Regression

Univariate: Single feature X = (x)

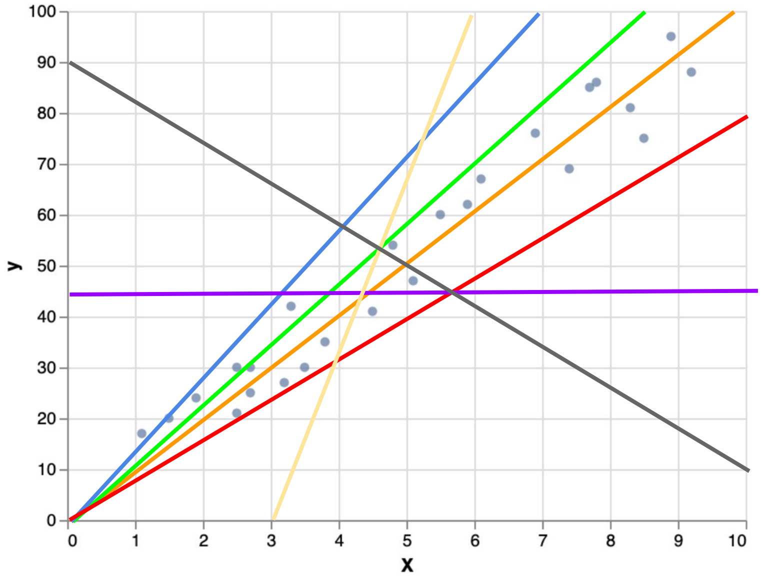
Linear: Mathematical relationship found between target (y) and feature (X) is a straight line

Regression: Target is continuous (infinite possible value within a range)



Algorithm Parameters

Algorithm Parameters: it is an internal parameter of the Machine Learning model that is automatically learned during the training on the data



Univariate Linear Regression:

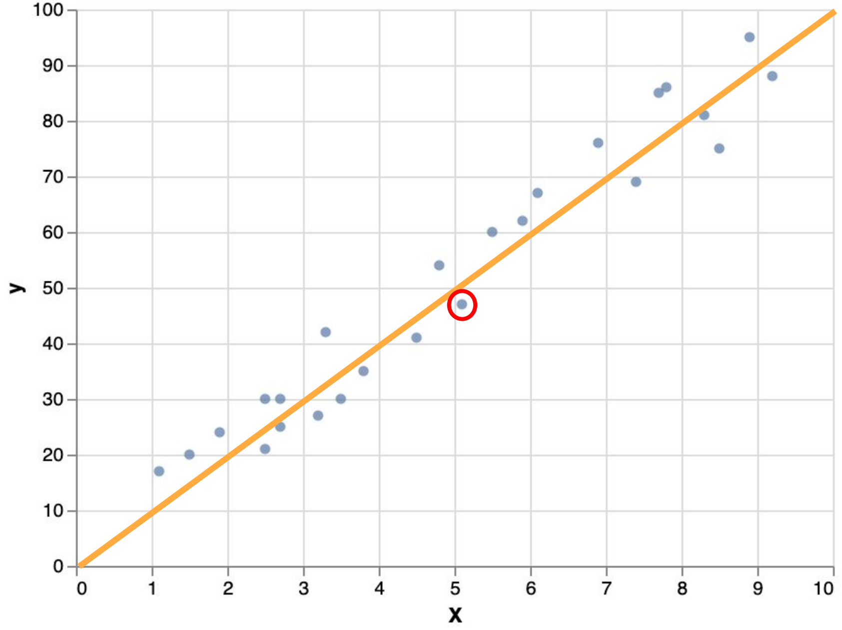
- Intercept: constant factor that corresponds to the expected value of y when X=0
- Coefficient: slope of the regression line (defines its steepness)

$$f(X) = w_0 + w_1 * x_1$$

intercept

coefficient

Fitting a Model



Univariate Linear Regression:

- Intercept: constant factor that corresponds to the expected value of y when X=0
- Coefficient: slope of the regression line (defines its steepness)

Parameters found:

$$f(X) = 0 + 10 * x_1 = 10 * x_1$$

Prediction:

$$\hat{y} = 10 * x = 10 * 5.1 = 51$$

Model Parameters vs Hyperparameters

Parameters are the configuration model, which are internal to the model. Their values are defined by the model (learning process).

Parameters are essential for making predictions.

Parameter	Hyperparameter
Estimated during the training with historical data.	Values are set beforehand.
It is a part of the model.	External to the model.
The estimated value is saved with the trained model.	Not a part of the trained model and hence the values are not saved.
Dependent on the dataset that the system is trained with.	Independent of the dataset.

Hyperparameters are the explicitly specified parameters (by data scientists) that control the training process (speed, performance, generalisation…).

Hyperparameters are essential for optimizing the model.

Sklearn "Parameters" (Hyperparameters)

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html#sklearn.linear_model.LinearRegression.set_params

Parameters:

fit_intercept : *bool, default=True*

Whether to calculate the intercept for this model. If set to False, no intercept will be used in calculations (i.e. data is expected to be centered).

normalize : *bool, default=False*

This parameter is ignored when `fit_intercept` is set to False. If True, the regressors X will be normalized before regression by subtracting the mean and dividing by the l2-norm. If you wish to standardize, please use `sklearn.preprocessing.StandardScaler` before calling `fit` on an estimator with `normalize=False`.

copy_X : *bool, default=True*

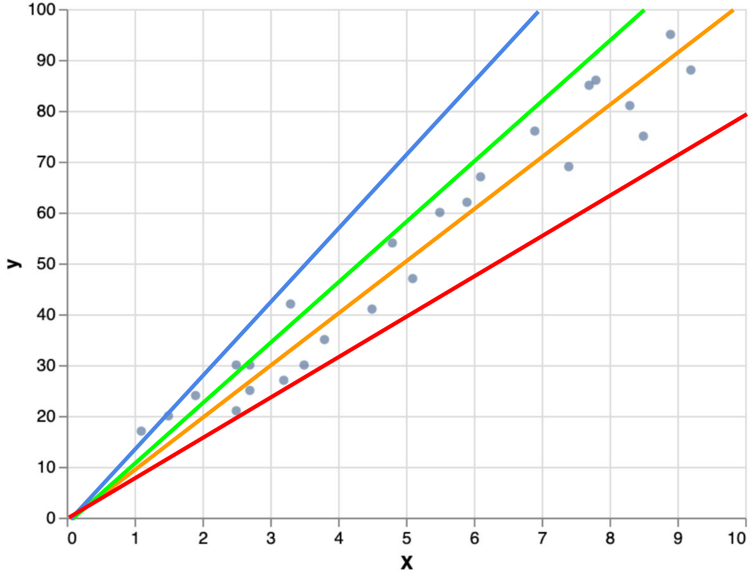
If True, X will be copied; else, it may be overwritten.

n_jobs : *int, default=None*

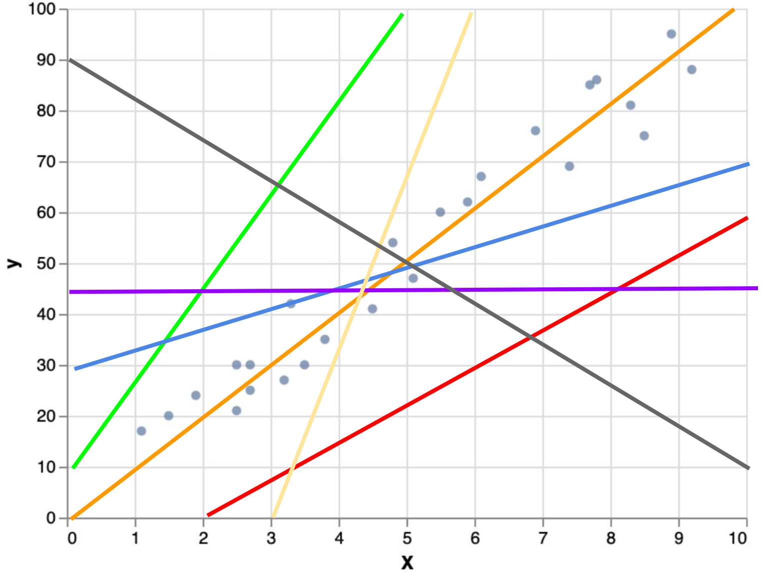
The number of jobs to use for the computation. This will only provide speedup for n_targets > 1 and sufficient large problems. `None` means 1 unless in a `joblib.parallel_backend` context. `-1` means using all processors. See Glossary for more details.

Algorithm Hyperparameters: it is a parameter external to the Machine Learning model. It is set by data scientists. This parameter influences the learning process of the model (speed, performance, generalisation…)

LR Hyperparameters



fit_intercept = False



fit_intercept = True

Scikit-Learn Package

- Open source Python package
- Initially launched in 2007
- Provides implementation of main ML algorithms

Benefits:

- Simple and Consistent API:
    - A uniform API for various ML models, making it easy to switch between algorithms.
- Abundance of Algorithms:
    - Offers a rich selection of supervised and unsupervised learning algorithms.
- Data Preprocessing:
    - Tools for preprocessing, feature selection, and dimensionality reduction.

Sklearn API

Importing scikit-learn:

- `import sklearn`

Data Loading:

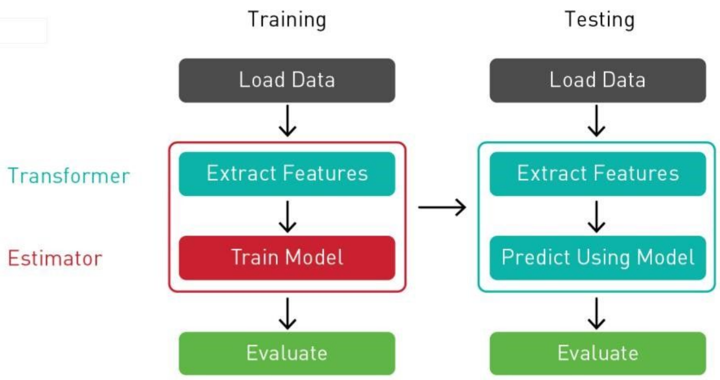
- Loading datasets with `datasets` module.

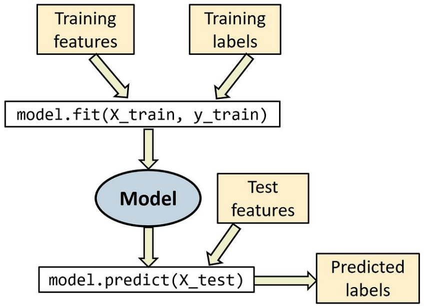
Creating a Model:

- Initialize a model, set hyperparameters, and fit to data.

Evaluation:

- Assess model performance using metrics from `metrics` module.





Example: Linear Regression

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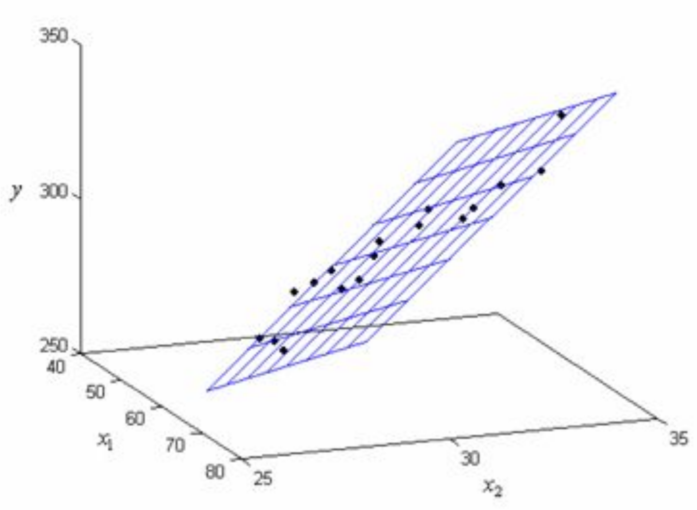
Regression: Target is continuous (infinite possible value within a range)



$$f(X) = f(x_1, x_2, \dots, x_n) = w_0 + w_1 * x_1 + w_2 * x_2 + \dots + w_n * x_n = \hat{y}$$

Scikit-learn Multivariate Linear Regression

Generic API for sklearn models:

- Import class

```
from sklearn.linear_model import LinearRegression
```

- Instantiate a model with hyperparameter

```
reg = LinearRegression()
```

- Fit a model

```
reg = reg.fit(X, y)
```

- Predict with a model

```
reg.predict(new_X)
```

Time for Lab

51